Table S1. Additional FinnGen GD screen discoveries

Gene	Chr	FinnGen GD IVs	FinnGen GD OR (95% CI)	FinnGen GD *P*	BBJ PP.H4	FinnGen GD PP.H4	FinnGen GO strict PP.H4
APOM	6 (MHC)	1	0.05 (0.04–0.06)	1.43×10^{-108}	<0.001	<0.001	0.227
NEU1	6 (MHC)	1	0.04 (0.03–0.06)	1.43×10^{-108}	<0.001	<0.001	0.061
IGLV2-18	22	1	0.56 (0.48–0.66)	7.52×10^{-12}	0.029	0.020	0.100
IGLV3-12	22	1	1.20 (1.14–1.26)	7.52×10^{-12}	0.019	0.022	0.101
TNF	6 (MHC)	1	0.60 (0.51–0.71)	4.65×10^{-10}	<0.001	<0.001	<0.001
HIST1H4E	6 (MHC)	1	2.54 (1.88–3.43)	1.51×10^{-9}	<0.001	<0.001	0.007
LTA	6 (MHC)	1	0.46 (0.35–0.61)	3.75×10^{-8}	<0.001	<0.001	<0.001
LTB	6 (MHC)	1	0.18 (0.10–0.36)	4.89×10^{-7}	<0.001	<0.001	<0.001
C4B	6 (MHC)	2	1.63 (1.31–2.03)	1.10×10^{-5}	<0.001	<0.001	0.311

Genes outside the BBJ discovery set that passed $P < 0.05/2{,}544$ in the post hoc full FinnGen GD screen (1000 Genomes European reference frequencies). These are not independent of the screen and seven of nine are in the MHC. *APOM*/*NEU1* share rs3101018 and *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. None reached PP.H4 ≥0.80 ($p12 = 10^{-5}$) in any dataset. Chr, chromosome; IV, instrumental variable; MHC, major histocompatibility complex; OR, odds ratio; PP.H4, shared-variant posterior probability; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.